Module 12: Darwin and Evolution - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Explain natural selection using variation, heritability, and reproductive success.
2. Define fitness in evolutionary terms.
3. Distinguish individual change from population evolution.
4. Use multiple evidence types to support common ancestry.
5. Apply natural selection to a concrete trait example.

Topics

- Historical explanation for adaptation
- Variation and heritability
- Natural selection mechanism
- Evidence for common ancestry
- Fitness as reproductive success

Key Terms to Know

- **Evolution** - Change in heritable traits of populations across generations.
- **Natural selection** - Nonrandom differences in survival or reproduction tied to heritable traits.
- **Adaptation** - Heritable trait shaped by selection in a context.
- **Fitness** - Relative reproductive success.
- **Common ancestry** - Shared origin of lineages through descent.
- **Homology** - Similarity due to shared ancestry.
- **Fossil** - Preserved evidence of past life.
- **Biogeography** - Study of geographic patterns in life.

Core Contents

1. Darwinian evolution explains adaptation through descent with modification.
2. Natural selection requires variation, heritability, and unequal reproductive success.
3. Fitness means relative reproductive success in a particular environment.
4. Fossils, homology, biogeography, and molecular evidence support common ancestry.
5. Evolution changes populations over generations, not individual organisms during life.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the module's companion lab (see the course schedule pairing) as the evidence check.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-13_darwin-evolution.md`

Generated Visuals

- Module 12: Darwin and Evolution Evidence Map
- Module 12: Darwin and Evolution Reasoning Sequence
- Module 12: Darwin and Evolution Retrieval and Lab Check